

MAIE 6000C: Engineering AI Products: From Prototype to Production

Week 2 – Lab

Program: MSc in AI & Entrepreneurship

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Session 1 divider



Session 1 — From Project Brief to First System Plan



Goal: narrow one project idea into a concrete, buildable workflow



By the end of this session, your team will have a project brief, problem statement, primary user, trigger event, and one end-to-end workflow.

Welcome and lab goal

Today's lab goal

By the end of today, your team will have:

- One candidate project direction
- One draft problem statement
- One primary user and trigger event
- One core workflow
- Initial requirements and decomposition notes
- A realistic Week 7 vertical-slice idea
- Initial risks and scope cuts

What we are producing today

- **Required outputs**
- Project brief
- Problem statement
- Primary user and trigger event
- Core workflow
- Functional requirements
- Initial entities and lifecycle states
- Bounded AI-enabled function and fallback
- Week 7 vertical slice
- Top three risks and first three scope cuts

Sketch only

- Non-functional requirements
- First API or service boundary
- Possible worker path
- Core components
- Mapping to the starter architecture

Task: Form teams or provisional groups

Activity — 5 minutes

- Work in existing teams or form provisional groups
- Identify one shared document or workspace (Proposed: Microsoft Word Shared Document)
- Assign a note-taker and discussion facilitator
- Aim for a realistic, buildable direction

Do not spend today resolving final team membership.



Starting point: Week 1 note or new brief

Choose your starting point

- Continue from a Week 1 candidate direction
- Or compare 1–2 new briefs today
- Or propose a new brief that fulfils the requirements of the proposed briefs (Primary users, Core Workflow, Possible AI Function)

Avoid starting from:

- “AI for everything”
- “Smart platform”
- “Chatbot for all tasks”
- Multiple unrelated user groups
- Multiple workflows at once



Task: Review project briefs

Review up to two possible briefs **or Continue from a Week 1 candidate direction **or** propose a new brief**

— 7 minutes

- For each brief, answer:
- Who is the primary user?
- What triggers the workflow?
- What enters the system?
- What must be remembered?
- What is the bounded AI function?
- What could be demonstrated by Week 7?
- Compare buildability, not novelty (check next slide's buildability checklist).



Required tool

- Microsoft Word shared document

Buildability questions



Buildability checklist

- Is there one clear primary user?
- Is there one concrete first workflow?
- Is there one clear input or trigger?
- Does the system need to remember a record or status?
- Is the AI function narrow and testable?
- Is there a review or fallback path?
- Can a small version be shown by Week 7?
- What will we deliberately not build?

If the answer to several questions is unclear, narrow the idea before continuing.

Task: Choose one candidate brief for today

Choose one candidate — 5 minutes

Write:

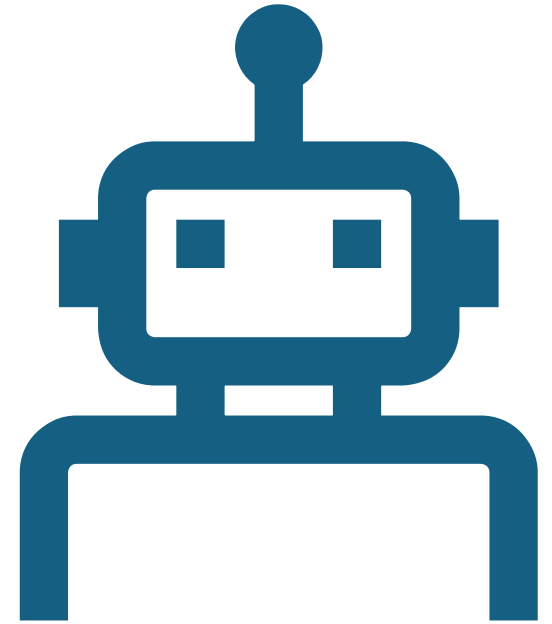
Our project is about...

- It is buildable because...
- We will not build...
- Our smallest Week 7 demo is...

Required tool

- Microsoft Word shared document

This is a working decision for today, not an irreversible final commitment.



Task: Draft the problem statement – 8 minutes

For **[primary user]**, the system helps **[user task]** by **[specific workflow]**, so that **[operational value]**.

Then add:

- Operational context
- Trigger event
- What success looks like
- One or two scope exclusions

Required tool

- Microsoft Word shared document



Suggested document structure

Project title:

Problem statement:

Primary user:

Operational context:

Trigger event:

What success looks like:

Out of scope:

- 1.
- 2.

Check Weak vs. Strong Problem Statement
Examples on the next slide

Task: Identify primary user and trigger event – 6 minutes

Fill in these fields

- Primary user:
- Trigger event:
- System input:
- First system output or decision:
- Who notices first if the AI is wrong?
- One thing the system will not handle:

Required tool

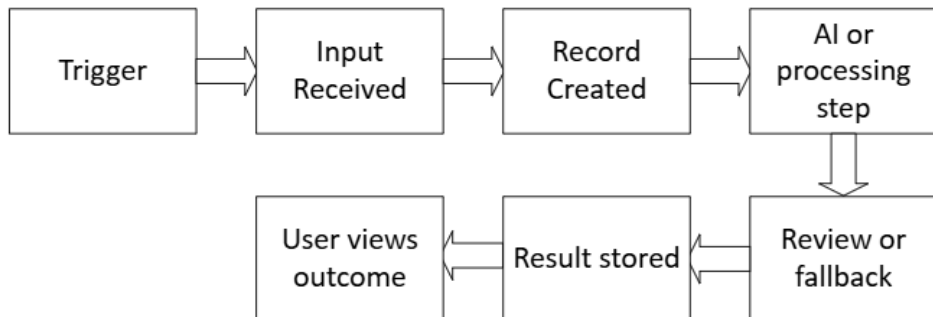
- Microsoft Word shared document



a homeowner interacting with a smart home system

Task: Draft the core workflow – 8 minutes

Recommended workflow format



Draft one complete workflow from trigger to result:

- A user or external event provides an input
- The system records or accepts the input
- The system processes or queues the work
- A bounded AI function produces an output
- A person or rule reviews the result when necessary
- The result and status are stored
- The user views or retrieves the outcome

Required Tool:

- Shared Microsoft Visio Drawing



Session 1 checkpoint

By now, each group should have

- One candidate project brief
- One problem statement
- One primary user
- One trigger event
- One core workflow
- At least two scope exclusions

Session 2 — Requirements and Decomposition

Session goal:

By the end of Session 2, each team should have:

- requirements
- entities
- components
- a Week 7 vertical slice
- risks and scope cuts

Restart and checkpoint – 4 minutes

Quickly confirm that the team has:

- A project brief
- A problem statement
- A primary user
- A trigger event
- A core workflow
- At least two scope exclusions

Required tool

- The same shared Microsoft Word document

Session 1 item	Complete?	Needs revision?
Project brief		
Problem statement		
Primary user		
Trigger event		
Core workflow		
Scope exclusions		

***If your team is missing one item, fix it now.
Otherwise, move directly to requirements.***

Task: Functional requirements – 9 minutes



Write **five** functional requirements based on the Session 1 workflow.



Start with **verbs** such as: create / accept / store / process / classify / retrieve / display / approve / override / notify / persist



Required tool:
Use a requirements table in the shared Microsoft Word document.

Requirements Table

ID	Functional requirement	Related workflow step	Observable evidence
F1	The system shall...		
F2	The system shall...		
F3	The system shall...		
F4	The system shall...		
F5	The system shall...		



Task: Non-functional requirements

– 4 minutes

Choose **three** quality priorities for the project

Possible options: reproducibility / reliability / maintainability / auditability / privacy / latency / cost / operability

For each one, write one sentence explaining why it matters.

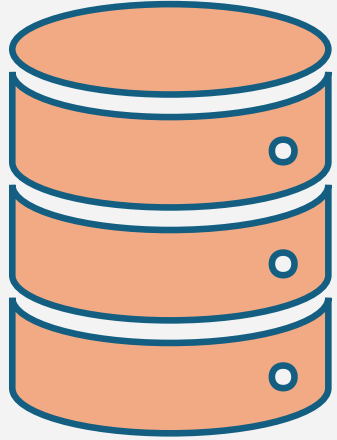
Required tool:

Use a three-row table in the shared Microsoft Word document.

Non-functional requirements table example

Quality priority	Why it matters	Initial implication
Auditability	Staff need to understand how a result was produced.	Store input, output, status, and review decision.
Reliability	Users must know whether work completed successfully.	Show processing status and failure state.
Privacy	Submitted reports may contain sensitive information.	Use anonymized data for the prototype.

You are identifying priorities, not setting final performance targets. Detailed measurements will come later in the course.



Task: Persistent entities or candidate records – 5 minutes

Identify the records the system must remember.

For each record, write:

- Record name
- Purpose
- Two or three likely attributes

Candidate record	Purpose	Likely attributes
Submitted item	Stores the original input	ID, content, submitter
Processing job	Tracks work being performed	ID, status, timestamps
AI result	Stores the generated output	category, score, explanation
Review decision	Stores human review	reviewer, decision, comment



Required tool

- Use a **candidate-records table** in the **shared Microsoft Word document**

Do not create SQL tables, field types, normalization rules, or migration files. These are candidate records only.



Task: Map your brief to the starter architecture - 2 minutes

Identify the approximate equivalent of:

- Submitted item or case
- Job
- Worker
- AI function
- Review
- Result

Required tool

- Use a mapping table in the shared Microsoft Word document

Project concept	Starter-architecture equivalent

Do NOT:

- Clone the repository
- Modify files
- Create branches
- Commit code
- Configure services

This is only a vocabulary-mapping exercise. You are connecting your project idea to the course architecture, not implementing it.



Task: First system boundary – 2 minutes

Answer four questions:

1. What is the first input?
2. What receives it?
3. What must happen immediately?
4. What can happen later?

Required tool

- Use a **four-row table** in the shared Microsoft Word document

Question	Team answer
First input	
Initial receiver	
Immediate action	
Later action	

If you already have a workflow diagram on Microsoft Visio Drawing, you can draw a dotted line around the initial receiver and the immediate action



Task: First worker path – 3 minutes

Complete:

- The background work is:
- It starts when:
- It reads:
- It writes:
- If it fails:

Required tool

- Use a **four-row table** in the shared **Microsoft Word document**

Example

Item	Team answer
Background work	Classify a submitted incident report
It starts when	A new report is accepted
It reads	Report text and record ID
It writes	Classification, confidence, and status
If it fails	Mark the job as failed and make it available for review

Task: First AI-enabled function and fallback

– 3 minutes

Complete:

- Given **[input]**, the system produces **[bounded output]** for **[user or workflow decision]**.

Then complete:

- If the result is unavailable, low-confidence, or invalid, **[fallback action]**.

Required tool

- Use an **AI function table** in the shared **Microsoft Word Document**

Example of AI function table

Item	Team answer
Input	Submitted incident report
Bounded AI Function	Classify the incident type
Output	One category and confidence score
User or decision supported	Staff prioritizes review
Fallback	Send low-confidence cases to manual review

Do not choose a final model, provider, prompt, training process, or evaluation metric today.

Task: Core components and responsibilities – 3 minutes

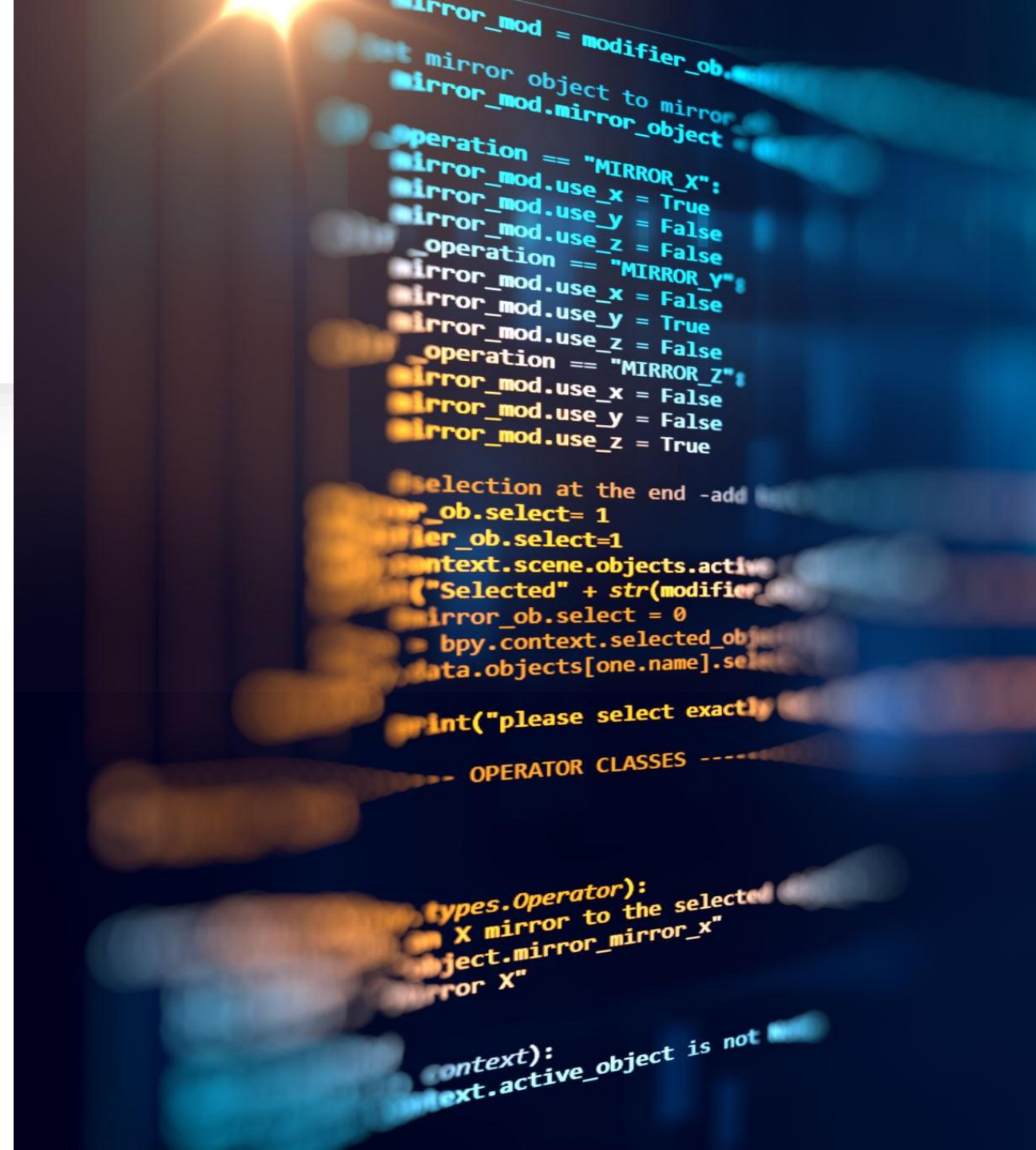
List the main components and give each one responsibility.

Required tool

- Use a **component table** in the shared Microsoft Word document.

Component	Responsibility
API or application service	Accept input and return status
Database	Store records and results
Worker	Perform background processing
AI module or service	Produce the bounded output
Review interface or endpoint	Allow human review
Tests and health checks	Provide basic operational evidence

You may remove components that do not apply.



Task: Week 7 vertical slice – 6 minutes

Complete:

- By Week 7, we will demonstrate...

Required tool

- Use a vertical-slice checklist in the shared Microsoft Word document.

Example

Required element	Example
Input	Staff uploads one incident report
Persisted state change	Report status changes to submitted
Processing step	Worker classifies the report
Persisted result	Category and confidence are stored
Operational signal	Status endpoint or log shows completion

Task: Top 3 risks and first 3 scope cuts – 5 minutes

Identify:

- Three major risks
- One mitigation or early check for each risk
- Three deliberate scope cuts

Required tool

- Use a **risk and scope table** in the **shared Microsoft Word Doc**.

A scope cut is not a weakness. It is evidence that the team understands what must be postponed.

Recommended format

Risk	Why it matters	Early check or mitigation
Data access	We may not have suitable input data	Use anonymized or synthetic examples
Model uncertainty	AI output may be unreliable	Add manual review and a simple baseline
Too many workflows	The team may overbuild	Implement only one workflow for Week 7

Then add:

Scope cut	Why it is acceptable
One user type	Keeps the workflow focused
One input type	Simplifies testing
Manual review	Avoids building complex automation



Task: Deliverable template and wrap-up – 2 minutes

Use a checklist to confirm that the shared planning document contains all required outputs.

Required tool

- Use a **checkbox checklist** in the shared Microsoft Word Document.

Required Session 2 content

- Five functional requirements
- Three non-functional requirements
- Candidate records/entities
- Conceptual lifecycle
- Bounded AI function
- Fallback path
- Week 7 vertical slice
- Three risks
- Three scope cuts